

Experiment 'p_2_8_rectangle_properties' Results

June 4, 2026

Experiment outcome: FAILURE_VERIFY

Bad responses: 1

Responses containing assume : 0

Responses containing expect : 0

Resolution attempts: 5

Hard fails (resolution): 0

Soft fails (resolution): 1

Verification attempts: 4

Problem Specification

Problem name: p_2_8_rectangle_properties

Natural language statement: null

Method signature: p_2_8_rectangle_properties(length: real, width: real) returns (area: real, perimeter: real, diagonal: real)

Ensures

- $\text{area} == \text{length} * \text{width}$
- $\text{perimeter} == 2.0 * (\text{length} + \text{width})$
- $-0.01 < (\text{diagonal} * \text{diagonal}) - ((\text{length} * \text{length}) + (\text{width} * \text{width})) < 0.01$

GenAI interactions

Below you will find all interactions between the ‘user’ (program) and the ‘assistant’ (OpenAI).

Program → GenAI

You are given the following task to perform in Dafny:

Write a method to satisfy the below requirements.

The signature should be:

`p_2_8_rectangle_properties(length: real, width: real) returns (area: real, perimeter: real, diagonal: real)`

The method should respect the following contract:

`ensures area == length * width, ensures perimeter == 2.0 * (length + width), ensures -0.01 < (diagonal * diagonal) - ((length * length) + (width * width)) < 0.01`

Produce and show only the Dafny body of this method, including the curly braces that surround it. Do not show the signature nor contract. You must not use 'assume' nor 'expect' anywhere in your code.

GenAI → Program

System fingerprint: null

ID: chatcmpl-Dn2nSYUgpWYvonnarnkGzY9jcyxjb

Model: gpt-5.5-2026-04-23

Created at: 1780581194

Finish reason: stop

Usage: 354 tokens in, and 1624 tokens out

```
{
  area := length * width;
  perimeter := 2.0 * (length + width);
  diagonal :| -0.01 < (diagonal * diagonal) - ((length * length) + (width * width)) < 0.01;
}
```

Program → GenAI

When using dafny verify, the below error is emitted and verify fails:

problem_data/p_2_8_rectangle_properties/program.dfy(8,11): Error: cannot establish the existence of LHS values that satisfy the such-that predicate. Note, no trigger was found for the such-that predicate, which may be the reason the proof failed. To give a trigger explicitly, use the { :trigger } attribute. For more information, see the section on quantifier instantiation rules in the reference manual.

```

8 | diagonal :| -0.01 < (diagonal * diagonal) - ((length * length) + (width * width)) <
    ↪ 0.01;
    | ^^

```

Dafny program verifier finished with 0 verified, 1 error

Correct the error in code by altering only the method body. Produce and show only the
 ↪ Dafny body, including the curly braces that surround it. Do not show the signature
 ↪ nor contract. You must not use 'assume' nor 'expect' anywhere in your code.

GenAI → Program

System fingerprint: null
ID: chatcmpl-Dn2oqFHQ6AU87WwpS3A9XQBvUyrrb
Model: gpt-5.5-2026-04-23
Created at: 1780581280
Finish reason: stop
Usage: 696 tokens in, and 3731 tokens out

```

{
  area := length * width;
  perimeter := 2.0 * (length + width);

  var s := (length * length) + (width * width);
  assert 0.0 <= s;
  assert exists d: real :: d * d == s;
  diagonal :| diagonal * diagonal == s;
  assert (diagonal * diagonal) - s == 0.0;
}

```

Program → GenAI

When using dafny resolve, the below error is emitted and resolve fails:

```

problem_data/p_2_8_rectangle_properties/program.dfy(11,9): Warning: Could not find a
    ↪ trigger for this quantifier. Without a trigger, the quantifier may cause brittle
    ↪ verification. To silence this warning, add an explicit trigger using the {:trigger
    ↪ } attribute. For more information, see the section on quantifier instantiation
    ↪ rules in the reference manual.
    |
11 | assert exists d: real :: d * d == s;
    | ~~~~~

```

Compilation failed because warnings were found and --allow-warnings is false

Correct the error in code by altering only the method body. Produce and show only the
 ↪ Dafny body, including the curly braces that surround it. Do not show the signature
 ↪ nor contract. You must not use 'assume' nor 'expect' anywhere in your code.

GenAI → Program

System fingerprint: null

ID: chatcmpl-Dn2qwpZ8AzAgJoxnkXt7Pbur7eDip

Model: gpt-5.5-2026-04-23

Created at: 1780581410

Finish reason: stop

Usage: 958 tokens in, and 4805 tokens out

```
{
  area := length * width;
  perimeter := 2.0 * (length + width);

  var s := (length * length) + (width * width);
  assert 0.0 <= s;
  assert exists d: real {:trigger d in {d}} :: d in {d} && d * d == s;
  diagonal :| diagonal in {diagonal} && diagonal * diagonal == s;
  assert (diagonal * diagonal) - s == 0.0;
  assert -0.01 < 0.0 < 0.01;
}
```

Program → GenAI

When using dafny verify, the below error is emitted and verify fails:

```
Encountered internal compilation exception: Boogie program had 1 resolution errors:
Microsoft.Dafny.SourceOrigin: equality is not allowed in triggers
problem_data/p_2_8_rectangle_properties/program.dfy(1,7): Error: Internal error occurred
  ↳ during verification: Boogie program had 1 resolution errors:
Microsoft.Dafny.SourceOrigin: equality is not allowed in triggers
  at Microsoft.Boogie.ExecutionEngine.GetVerificationTasks(Program program,
    ↳ CancellationTokentoken cancellationToken)
  at Microsoft.Dafny.LanguageServer.Language.DafnyProgramVerifier.
    ↳ GetVerificationTasksAsync(ExecutionEngine engine, ResolutionResult resolution,
    ↳ ModuleDefinition moduleDefinition, CancellationTokentoken cancellationToken)
  at Microsoft.Dafny.Compilation.<>c__DisplayClass60_0.<<VerifyUnverifiedSymbol>b__1>d.
    ↳ MoveNext()
--- End of stack trace from previous location ---
  at Microsoft.Dafny.Compilation.VerifyUnverifiedSymbol(Boolean
    ↳ onlyPrepareVerificationForGutterTests, ICanVerify canVerify, ResolutionResult
    ↳ resolution, Func`2 taskFilter, Nullable`1 randomSeed)
  at DafnyDriver.Commands.CliCompilation.VerifyAllLazily(Nullable`1 randomSeed)+MoveNext
    ↳ ()
|
1 | method p_2_8_rectangle_properties(length: real, width: real) returns (area: real,
    ↳ perimeter: real, diagonal: real)
| ~~~~~
Unhandled exception. System.ArgumentException: Boogie program had 1 resolution errors:
Microsoft.Dafny.SourceOrigin: equality is not allowed in triggers
  at Microsoft.Boogie.ExecutionEngine.GetVerificationTasks(Program program,
    ↳ CancellationTokentoken cancellationToken)
```

```

at Microsoft.Dafny.LanguageServer.Language.DafnyProgramVerifier.
  ↳ GetVerificationTasksAsync(ExecutionEngine engine, ResolutionResult resolution,
  ↳ ModuleDefinition moduleDefinition, CancellationToken cancellationToken)
at Microsoft.Dafny.Compilation.<>c__DisplayClass60_0.<<VerifyUnverifiedSymbol>b__1>d.
  ↳ MoveNext()
--- End of stack trace from previous location ---
at Microsoft.Dafny.Compilation.VerifyUnverifiedSymbol(Boolean
  ↳ onlyPrepareVerificationForGutterTests, ICanVerify canVerify, ResolutionResult
  ↳ resolution, Func`2 taskFilter, Nullable`1 randomSeed)
at DafnyDriver.Commands.CliCompilation.VerifyAllLazily(Nullable`1 randomSeed)+MoveNext
  ↳ ()
at DafnyDriver.Commands.CliCompilation.VerifyAllLazily(Nullable`1 randomSeed)+System.
  ↳ Threading.Tasks.Sources.IValueTaskSource<System.Boolean>.GetResult()
at System.Linq.AsyncEnumerable.ToObservableObservable`1.<>c__DisplayClass2_0.<<
  ↳ Subscribe>g__Core|0>d.MoveNext() in /_/Ix.NET/Source/System.Linq.Async/System/
  ↳ Linq/Operators/ToObservable.cs:line 50
--- End of stack trace from previous location ---
at System.Reactive.PlatformServices.ExceptionServicesImpl.Rethrow(Exception exception)
  ↳ in /_/Rx.NET/Source/src/System.Reactive/Internal/ExceptionServicesImpl.cs:line
  ↳ 16
at System.Reactive.ExceptionHelpers.Throw(Exception exception) in /_/Rx.NET/Source/src
  ↳ /System.Reactive/Internal/ExceptionServices.cs:line 14
at System.Reactive.Stubs.<>c.<.cctor>b__2_1(Exception ex) in /_/Rx.NET/Source/src/
  ↳ System.Reactive/Internal/Stubs.cs:line 16
at System.Reactive.AnonymousObserver`1.OnErrorCore(Exception error) in /_/Rx.NET/
  ↳ Source/src/System.Reactive/AnonymousObserver.cs:line 73
at System.Reactive.ObserverBase`1.OnError(Exception error) in /_/Rx.NET/Source/src/
  ↳ System.Reactive/ObserverBase.cs:line 59
at System.Reactive.Subjects.Subject`1.OnError(Exception error) in /_/Rx.NET/Source/src
  ↳ /System.Reactive/Subjects/Subject.cs:line 103
at System.Linq.AsyncEnumerable.ToObservableObservable`1.<>c__DisplayClass2_0.<<
  ↳ Subscribe>g__Core|0>d.MoveNext() in /_/Ix.NET/Source/System.Linq.Async/System/
  ↳ Linq/Operators/ToObservable.cs:line 60
--- End of stack trace from previous location ---
at System.Linq.AsyncEnumerable.ToObservableObservable`1.<>c__DisplayClass2_0.<<
  ↳ Subscribe>g__Core|0>d.MoveNext() in /_/Ix.NET/Source/System.Linq.Async/System/
  ↳ Linq/Operators/ToObservable.cs:line 74
--- End of stack trace from previous location ---
at System.Threading.Tasks.Task.<>c.<ThrowAsync>b__128_1(Object state)
at System.Threading.ThreadPoolWorkQueue.Dispatch()
at System.Threading.PortableThreadPool.WorkerThread.WorkerThreadStart()

Correct the error in code by altering only the method body. Produce and show only the
  ↳ Dafny body, including the curly braces that surround it. Do not show the signature
  ↳ nor contract. You must not use 'assume' nor 'expect' anywhere in your code.

```

GenAI → Program

System fingerprint: null

ID: chatcmpl-Dn2tluv3QsrkHMakY3BMaS4SDypKx

Model: gpt-5.5-2026-04-23

Created at: 1780581585

Finish reason: stop

Usage: 3011 tokens in, and 3798 tokens out

```
{
  area := length * width;
  perimeter := 2.0 * (length + width);

  var s := (length * length) + (width * width);
  assert 0.0 <= length * length;
  assert 0.0 <= width * width;
  assert 0.0 <= s;

  var triggerSet: set<real> := {};
  assert exists d: real {:trigger d in triggerSet} :: (d in triggerSet || !(d in
    ↪ triggerSet)) && d * d == s;
  diagonal :| (diagonal in triggerSet || !(diagonal in triggerSet)) && diagonal *
    ↪ diagonal == s;

  assert diagonal * diagonal == s;
  assert (diagonal * diagonal) - s == 0.0;
  assert -0.01 < 0.0 < 0.01;
}
```

Program → GenAI

When using dafny verify, the below error is emitted and verify fails:

```
problem_data/p_2_8_rectangle_properties/program.dfy(15,2): Error: assertion might not
  ↪ hold
|
```

```
15 | assert exists d: real {:trigger d in triggerSet} :: (d in triggerSet || !(d in
  ↪ triggerSet)) && d * d == s;
| ~~~~~
```

```
problem_data/p_2_8_rectangle_properties/program.dfy(16,11): Error: cannot establish the
  ↪ existence of LHS values that satisfy the such-that predicate
|
```

```
16 | diagonal :| (diagonal in triggerSet || !(diagonal in triggerSet)) && diagonal *
  ↪ diagonal == s;
| ^^
```

Dafny program verifier finished with 0 verified, 2 errors

Correct the error in code by altering only the method body. Produce and show only the
↪ Dafny body, including the curly braces that surround it. Do not show the signature
↪ nor contract. You must not use 'assume' nor 'expect' anywhere in your code.

GenAI → Program

System fingerprint: null

ID: chatcmpl-Dn2wBueZgTqYrGyXc6o4JsU42773E

Model: gpt-5.5-2026-04-23

Created at: 1780581735

Finish reason: stop

Usage: 2620 tokens in, and 1196 tokens out

```
{
  area := length * width;
  perimeter := 2.0 * (length + width);

  var s := length * length + width * width;
  assert 0.0 <= length * length;
  assert 0.0 <= width * width;
  assert 0.0 <= s;

  assert exists d: real {:trigger d * d} :: d * d == s;
  diagonal :| diagonal * diagonal == s;

  assert (diagonal * diagonal) - s == 0.0;
  assert -0.01 < 0.0;
  assert 0.0 < 0.01;
}
```

Program → GenAI

When using dafny verify, the below error is emitted and verify fails:

```
problem_data/p_2_8_rectangle_properties/program.dfy(14,2): Error: assertion might not
  ↪ hold
  |
14 | assert exists d: real {:trigger d * d} :: d * d == s;
  | ~~~~~
```

Dafny program verifier finished with 0 verified, 1 error

Correct the error in code by altering only the method body. Produce and show only the
 ↪ Dafny body, including the curly braces that surround it. Do not show the signature
 ↪ nor contract. You must not use 'assume' nor 'expect' anywhere in your code.

GenAI → Program

System fingerprint: null

ID: chatcmpl-Dn2xi0Zs53YifTELqDLaqA0sJpoWf

Model: gpt-5.5-2026-04-23

Created at: 1780581830

Finish reason: length

Usage: 2844 tokens in, and 7207 tokens out

The below is a raw response from GenAI that could not be parsed. It is likely that GenAI got 'stuck' in a loop and hit the pre-set token limit (i.e., finish reason is 'length').

Final Program

```
method p_2_8_rectangle_properties(length: real, width: real) returns (area: real,
  ↪ perimeter: real, diagonal: real)
  ensures area == length * width
  ensures perimeter == 2.0 * (length + width)
  ensures -0.01 < (diagonal * diagonal) - ((length * length) + (width * width)) <
    ↪ 0.01
{
  area := length * width;
  perimeter := 2.0 * (length + width);

  var s := length * length + width * width;
  assert 0.0 <= length * length;
  assert 0.0 <= width * width;
  assert 0.0 <= s;

  assert exists d: real {:trigger d * d} :: d * d == s;
  diagonal :| diagonal * diagonal == s;

  assert (diagonal * diagonal) - s == 0.0;
  assert -0.01 < 0.0;
  assert 0.0 < 0.01;
}
```

Total Token Usage

Input tokens: 10483

Output tokens: 22361

Reasoning tokens: 21514

Sum of 'total tokens': 32844

Experiment Timings

Overall Experiment started at 1780581192286, ended at 1780582016525, lasting 824239ms (824.24 seconds)

Iteration #4 started at 1780581582618, ended at 1780581734176, lasting 151558ms (151.56 seconds)

Iteration #5 started at 1780581734176, ended at 1780581823844, lasting 89668ms (89.67 seconds)

Iteration #6 started at 1780581823844, ended at 1780582016525, lasting 192681ms (192.68 seconds)

Iteration #1 started at 1780581192294, ended at 1780581279256, lasting 86962ms (86.96 seconds)

Iteration #2 started at 1780581279277, ended at 1780581408497, lasting 129220ms (129.22 seconds)

Iteration #3 started at 1780581408497, ended at 1780581582618, lasting 174121ms (174.12 seconds)